CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the textbook exercises: 28 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 8.1

Q1 In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Find (i) $\sin A$, $\cos A$ (ii) $\sin C$, $\cos C$.

SOLUTION

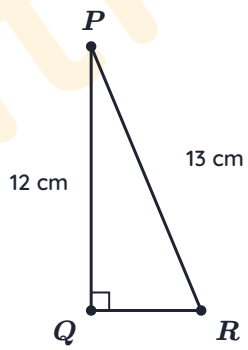
$AC = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$ cm.

(i) For angle A the opposite side is BC and the adjacent side is AB: $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$.

(ii) For angle C the opposite side is AB and the adjacent side is BC: $\sin C = \frac{24}{25}$, $\cos C = \frac{7}{25}$.

Answer: (i) $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25}$, $\cos C = \frac{7}{25}$

Q2 In $\triangle PQR$, right-angled at Q, $PQ = 12$ cm and $PR = 13$ cm (textbook Fig. 8.13). Find $\tan P - \cot R$.



SOLUTION

$QR = \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

$\tan P = \frac{QR}{PQ} = \frac{5}{12}$ and $\cot R = \frac{QR}{PQ} = \frac{5}{12}$.

Answer: $\tan P - \cot R = 0$

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Q3 If $\sin A = \frac{3}{4}$, find $\cos A$ and $\tan A$.

SOLUTION

Take the opposite side as $3k$ and the hypotenuse as $4k$. The adjacent side is $\sqrt{16k^2 - 9k^2} = \sqrt{7}\,k$.

Answer: $\cos A = \frac{\sqrt{7}}{4}$ **and** $\tan A = \frac{3}{\sqrt{7}}$

Q4 Given $15 \cot A = 8$, find $\sin A$ and $\sec A$.

SOLUTION

$\cot A = \frac{8}{15}$, so take the adjacent side as $8k$ and the opposite side as $15k$. The hypotenuse is $\sqrt{64k^2 + 225k^2} = 17k$.

Answer: $\sin A = \frac{15}{17}$ **and** $\sec A = \frac{17}{8}$

Q5 Given $\sec \theta = \frac{13}{12}$, find all the other trigonometric ratios.

SOLUTION

Take the hypotenuse as $13k$ and the adjacent side as $12k$. The opposite side is $\sqrt{169k^2 - 144k^2} = 5k$.

Answer: $\sin \theta = \frac{5}{13}$, $\cos \theta = \frac{12}{13}$, $\tan \theta = \frac{5}{12}$, $\operatorname{cosec} \theta = \frac{13}{5}$, $\cot \theta = \frac{12}{5}$

Q6 $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$. Show that $\angle A = \angle B$.

SOLUTION

Take a triangle ABC right-angled at C, so that $\angle A$ and $\angle B$ are its acute angles. Then $\cos A = \frac{AC}{AB}$ and $\cos B = \frac{BC}{AB}$.

$\cos A = \cos B$ gives $AC = BC$.

Angles opposite equal sides of a triangle are equal, so $\angle B = \angle A$.

Hence $\angle A = \angle B$.

Q7 If $\cot \theta = \frac{7}{8}$, evaluate (i) $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$ (ii) $\cot^2 \theta$.

SOLUTION

(i) $\frac{1 - \sin^2 \theta}{1 - \cos^2 \theta} = \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta = \frac{49}{64}$.

(ii) $\cot^2 \theta = \left(\frac{7}{8}\right)^2 = \frac{49}{64}$.

Answer: (i) $\frac{49}{64}$ **(ii)** $\frac{49}{64}$

Q8 If $3\cot A = 4$, check whether $\dfrac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$.

SOLUTION

$\cot A = \dfrac{4}{3}$, so the adjacent side is $4k$, the opposite side is $3k$ and the hypotenuse is $5k$: $\tan A = \dfrac{3}{4}$, $\sin A = \dfrac{3}{5}$, $\cos A = \dfrac{4}{5}$.

LHS $= \dfrac{1-\frac{9}{16}}{1+\frac{9}{16}} = \dfrac{7}{25}$. RHS $= \dfrac{16}{25} - \dfrac{9}{25} = \dfrac{7}{25}$.

Answer: yes, both sides equal $\dfrac{7}{25}$

Q9 In $\triangle ABC$, right-angled at B, $\tan A = \dfrac{1}{\sqrt{3}}$. Find (i) $\sin A\cos C + \cos A \sin C$ (ii) $\cos A \cos C - \sin A \sin C$.

SOLUTION

Take $BC = k$ and $AB = \sqrt{3}\,k$, so $AC = \sqrt{k^2 + 3k^2} = 2k$.

$\sin A = \dfrac{1}{2}$, $\cos A = \dfrac{\sqrt{3}}{2}$, $\sin C = \dfrac{\sqrt{3}}{2}$, $\cos C = \dfrac{1}{2}$.

(i) $\dfrac{1}{2}\times\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\times\dfrac{\sqrt{3}}{2} = \dfrac{1}{4} + \dfrac{3}{4} = 1$. (ii) $\dfrac{\sqrt{3}}{2}\times\dfrac{1}{2} - \dfrac{1}{2}\times\dfrac{\sqrt{3}}{2} = 0$.

Answer: (i) 1 (ii) 0

Q10 In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Find $\sin P$, $\cos P$ and $\tan P$.

SOLUTION

Let $QR = x$ cm, so $PR = (25 - x)$ cm. By the Pythagoras theorem, $(25-x)^2 = 5^2 + x^2$.

$625 - 50x = 25$ gives $x = 12$. So $QR = 12$ cm and $PR = 13$ cm.

Answer: $\sin P = \dfrac{12}{13}$, $\cos P = \dfrac{5}{13}$, $\tan P = \dfrac{12}{5}$

Q11 State whether each is true or false, with a reason. (i) The value of $\tan A$ is always less than 1. (ii) $\sec A = \dfrac{12}{5}$ for some value of angle A. (iii) $\cos A$ is the abbreviation for the cosecant of angle A. (iv) $\cot A$ is the product of cot and A. (v) $\sin\theta = \dfrac{4}{3}$ for some angle θ.

SOLUTION

(i) False. For example, $\tan 60^\circ = \sqrt{3} > 1$.

(ii) True. $\sec A$ is the hypotenuse divided by the adjacent side, which can be any number from 1 upwards, and $\dfrac{12}{5} > 1$.

(iii) False. $\cos A$ stands for the cosine of A. The cosecant is written $\operatorname{cosec} A$.

(iv) False. $\cot A$ is one symbol, the cotangent of the angle A. "cot" alone has no meaning.

(v) False. $\sin\theta$ is the opposite side divided by the hypotenuse, and the hypotenuse is the longest side, so $\sin\theta$ cannot be more than 1.

Answer: (i) False (ii) True (iii) False (iv) False (v) False

2 Exercise 8.2

Q1 Evaluate: (i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ (ii) $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ (iii) $\dfrac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$ (iv) $\dfrac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$ (v) $\dfrac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.

SOLUTION

(i) $\dfrac{\sqrt{3}}{2} \times \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{3}{4} + \dfrac{1}{4} = 1.$

(ii) $2(1)^2 + \dfrac{3}{4} - \dfrac{3}{4} = 2.$

(iii) $\dfrac{\frac{1}{\sqrt{2}}}{\frac{2}{\sqrt{3}} + 2} = \dfrac{\sqrt{3}}{2\sqrt{2}\,(\sqrt{3}+1)}$. Multiply the top and bottom by $(\sqrt{3}-1)$: $\dfrac{\sqrt{3}\,(\sqrt{3}-1)}{4\sqrt{2}} = \dfrac{3-\sqrt{3}}{4\sqrt{2}} = \dfrac{3\sqrt{2}-\sqrt{6}}{8}$.

(iv) $\dfrac{\frac{1}{2} + 1 - \frac{2}{\sqrt{3}}}{\frac{2}{\sqrt{3}} + \frac{1}{2} + 1} = \dfrac{3\sqrt{3}-4}{3\sqrt{3}+4}$. Multiply the top and bottom by $(3\sqrt{3}-4)$: $\dfrac{(3\sqrt{3}-4)^2}{27-16} = \dfrac{43-24\sqrt{3}}{11}$.

(v) $\dfrac{5 \times \frac{1}{4} + 4 \times \frac{4}{3} - 1}{\frac{1}{4} + \frac{3}{4}} = \dfrac{15 + 64 - 12}{12} = \dfrac{67}{12}$.

Answer: (i) 1 (ii) 2 (iii) $\dfrac{3\sqrt{2}-\sqrt{6}}{8}$ (iv) $\dfrac{43-24\sqrt{3}}{11}$ (v) $\dfrac{67}{12}$

Q2 Choose the correct option and justify it. (i) $\dfrac{2\tan 30^\circ}{1+\tan^2 30^\circ}$ = (A) $\sin 60^\circ$ (B) $\cos 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$.
(ii) $\dfrac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ = (A) $\tan 90^\circ$ (B) 1 (C) $\sin 45^\circ$ (D) 0. (iii) $\sin 2A = 2\sin A$ is true when A = (A) 0° (B) 30° (C) 45° (D) 60°. (iv) $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ = (A) $\cos 60^\circ$ (B) $\sin 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$.

SOLUTION

(i) $\dfrac{2 \times \frac{1}{\sqrt{3}}}{1 + \frac{1}{3}} = \dfrac{2}{\sqrt{3}} \times \dfrac{3}{4} = \dfrac{\sqrt{3}}{2} = \sin 60^\circ$. Option (A).

(ii) $\dfrac{1-1}{1+1} = 0$. Option (D).

(iii) For $A = 0^\circ$: $\sin 0^\circ = 0$ and $2\sin 0^\circ = 0$, so it holds. For the other options it fails, for example $\sin 60^\circ \neq 2\sin 30^\circ$. Option (A).

(iv) $\dfrac{2 \times \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}} = \dfrac{2}{\sqrt{3}} \times \dfrac{3}{2} = \sqrt{3} = \tan 60^\circ$. Option (C).

Answer: (i) A (ii) D (iii) A (iv) C

Q3 If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \dfrac{1}{\sqrt{3}}$, with $0^\circ < A+B \leq 90^\circ$ and $A > B$, find A and B.

SOLUTION

$\tan(A+B) = \sqrt{3} = \tan 60^\circ$ gives $A+B = 60^\circ$. $\tan(A-B) = \dfrac{1}{\sqrt{3}} = \tan 30^\circ$ gives $A-B = 30^\circ$.

Adding, $2A = 90^\circ$, so $A = 45^\circ$ and $B = 15^\circ$.

Answer: $A = 45^\circ$, $B = 15^\circ$

Q4 State whether each is true or false, with a reason. (i) $\sin(A+B) = \sin A + \sin B$. (ii) The value of $\sin\theta$ increases as θ increases. (iii) The value of $\cos\theta$ increases as θ increases. (iv) $\sin\theta = \cos\theta$ for all values of θ. (v) $\cot A$ is not defined for $A = 0°$.

SOLUTION

(i) False. With $A = 60°$, $B = 30°$: $\sin 90° = 1$ but $\sin 60° + \sin 30° = \dfrac{\sqrt{3}+1}{2} \neq 1$.

(ii) True for $0° \leq \theta \leq 90°$: the values go $0, \dfrac{1}{2}, \dfrac{1}{\sqrt{2}}, \dfrac{\sqrt{3}}{2}, 1$.

(iii) False. From $0°$ to $90°$ the values of $\cos\theta$ go from 1 down to 0.

(iv) False. They are equal only for $\theta = 45°$. For example $\sin 30° = \dfrac{1}{2}$ but $\cos 30° = \dfrac{\sqrt{3}}{2}$.

(v) True. $\cot 0° = \dfrac{\cos 0°}{\sin 0°} = \dfrac{1}{0}$, which is not defined.

Answer: (i) False (ii) True (iii) False (iv) False (v) True

3 Exercise 8.3

Q1 Express $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

SOLUTION

$\operatorname{cosec}^2 A = 1 + \cot^2 A$, so $\sin A = \dfrac{1}{\operatorname{cosec} A} = \dfrac{1}{\sqrt{1+\cot^2 A}}$.

$\tan A = \dfrac{1}{\cot A}$.

$\sec^2 A = 1 + \tan^2 A = 1 + \dfrac{1}{\cot^2 A}$, so $\sec A = \dfrac{\sqrt{1+\cot^2 A}}{\cot A}$.

Answer: $\sin A = \dfrac{1}{\sqrt{1+\cot^2 A}}$, $\sec A = \dfrac{\sqrt{1+\cot^2 A}}{\cot A}$, $\tan A = \dfrac{1}{\cot A}$

Q2 Write all the other trigonometric ratios of $\angle A$ in terms of $\sec A$.

SOLUTION

$\cos A = \dfrac{1}{\sec A}$.

$\tan^2 A = \sec^2 A - 1$, so $\tan A = \sqrt{\sec^2 A - 1}$ and $\cot A = \dfrac{1}{\sqrt{\sec^2 A - 1}}$.

$\sin A = \tan A \times \cos A = \dfrac{\sqrt{\sec^2 A - 1}}{\sec A}$ and $\operatorname{cosec} A = \dfrac{\sec A}{\sqrt{\sec^2 A - 1}}$.

Answer: $\cos A = \dfrac{1}{\sec A}$, $\sin A = \dfrac{\sqrt{\sec^2 A - 1}}{\sec A}$, $\tan A = \sqrt{\sec^2 A - 1}$, $\cot A = \dfrac{1}{\sqrt{\sec^2 A - 1}}$, $\operatorname{cosec} A = \dfrac{\sec A}{\sqrt{\sec^2 A - 1}}$

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Q3 Choose the correct option and justify it. (i) $9\sec^2 A - 9\tan^2 A =$ (A) 1 (B) 9 (C) 8 (D) 0. (ii) $(1+\tan\theta+\sec\theta)(1+\cot\theta-\operatorname{cosec}\theta) =$ (A) 0 (B) 1 (C) 2 (D) -1. (iii) $(\sec A+\tan A)(1-\sin A) =$ (A) $\sec A$ (B) $\sin A$ (C) $\operatorname{cosec} A$ (D) $\cos A$. (iv) $\dfrac{1+\tan^2 A}{1+\cot^2 A} =$ (A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$.

SOLUTION

(i) $9(\sec^2 A - \tan^2 A) = 9 \times 1 = 9$. Option (B).

(ii) $\dfrac{\cos\theta+\sin\theta+1}{\cos\theta} \times \dfrac{\sin\theta+\cos\theta-1}{\sin\theta} = \dfrac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta} = \dfrac{2\sin\theta\cos\theta}{\sin\theta\cos\theta} = 2$. Option (C).

(iii) $\dfrac{1+\sin A}{\cos A} \times (1-\sin A) = \dfrac{1-\sin^2 A}{\cos A} = \cos A$. Option (D).

(iv) $\dfrac{\sec^2 A}{\operatorname{cosec}^2 A} = \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$. Option (D).

Answer: (i) B (ii) C (iii) D (iv) D

Q4 (i) Prove that $(\operatorname{cosec}\theta - \cot\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$.

SOLUTION

$$\text{LHS} = \left(\frac{1-\cos\theta}{\sin\theta}\right)^2 = \frac{(1-\cos\theta)^2}{1-\cos^2\theta} = \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)}$$

$$= \frac{1-\cos\theta}{1+\cos\theta} = \text{RHS}$$

Q4 (ii) Prove that $\dfrac{\cos A}{1+\sin A} + \dfrac{1+\sin A}{\cos A} = 2\sec A$.

SOLUTION

$$\text{LHS} = \frac{\cos^2 A + (1+\sin A)^2}{(1+\sin A)\cos A} = \frac{\cos^2 A + \sin^2 A + 1 + 2\sin A}{(1+\sin A)\cos A} = \frac{2(1+\sin A)}{(1+\sin A)\cos A}$$

$$= \frac{2}{\cos A} = 2\sec A = \text{RHS}$$

Q4 (iii) Prove that $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\operatorname{cosec}\theta$.

SOLUTION

Write everything with $\sin\theta$ and $\cos\theta$: $\text{LHS} = \dfrac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \dfrac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)} = \dfrac{\sin^3\theta - \cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$

Using $a^3-b^3 = (a-b)(a^2+ab+b^2)$: $= \dfrac{\sin^2\theta+\sin\theta\cos\theta+\cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta}$

$= \sec\theta\operatorname{cosec}\theta + 1 = \text{RHS}$

Q4 (iv) Prove that $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$.

SOLUTION

LHS $= \frac{1}{\sec A} + 1 = \cos A + 1$.

RHS $= \frac{1-\cos^2 A}{1-\cos A} = \frac{(1-\cos A)(1+\cos A)}{1-\cos A} = 1+\cos A$.

LHS = RHS

Q4 (v) Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using $\operatorname{cosec}^2 A = 1 + \cot^2 A$.

SOLUTION

Divide the numerator and the denominator by $\sin A$: LHS $= \frac{\cot A - 1 + \operatorname{cosec} A}{\cot A + 1 - \operatorname{cosec} A}$.

In the numerator replace 1 by $\operatorname{cosec}^2 A - \cot^2 A$: $(\cot A + \operatorname{cosec} A) - (\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A) = (\cot A + \operatorname{cosec} A)(1 - \operatorname{cosec} A + \cot A)$.

So LHS $= \frac{(\cot A + \operatorname{cosec} A)(\cot A + 1 - \operatorname{cosec} A)}{\cot A + 1 - \operatorname{cosec} A} = \operatorname{cosec} A + \cot A =$ **RHS**

Q4 (vi) Prove that $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$.

SOLUTION

Multiply the numerator and the denominator inside the root by $(1+\sin A)$: LHS $= \sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}} = \frac{1+\sin A}{\cos A}$

$= \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \tan A =$ **RHS**

Q4 (vii) Prove that $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$.

SOLUTION

LHS $= \frac{\sin\theta\,(1-2\sin^2\theta)}{\cos\theta\,(2\cos^2\theta - 1)}$.

Using $1 = \sin^2\theta + \cos^2\theta$: $1 - 2\sin^2\theta = \cos^2\theta - \sin^2\theta$ and $2\cos^2\theta - 1 = \cos^2\theta - \sin^2\theta$.

So LHS $= \frac{\sin\theta}{\cos\theta} = \tan\theta =$ **RHS**

Q4 (viii) Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

SOLUTION

LHS $= \sin^2 A + \operatorname{cosec}^2 A + 2 + \cos^2 A + \sec^2 A + 2$, since $\sin A \times \operatorname{cosec} A = 1$ and $\cos A \times \sec A = 1$.

$= (\sin^2 A + \cos^2 A) + 4 + (1 + \cot^2 A) + (1 + \tan^2 A)$

$= 7 + \tan^2 A + \cot^2 A =$ **RHS**

Q4 (ix) Prove that $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$.

SOLUTION

$$\text{LHS} = \frac{1-\sin^2 A}{\sin A} \times \frac{1-\cos^2 A}{\cos A} = \frac{\cos^2 A \sin^2 A}{\sin A \cos A} = \sin A \cos A.$$

$$\text{RHS} = \frac{1}{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}} = \frac{\sin A \cos A}{\sin^2 A + \cos^2 A} = \sin A \cos A.$$

LHS = RHS

Q4 (x) Prove that $\dfrac{1+\tan^2 A}{1+\cot^2 A} = \left(\dfrac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$.

SOLUTION

First expression: $\dfrac{\sec^2 A}{\operatorname{cosec}^2 A} = \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$.

Second expression: $1 - \cot A = \dfrac{\tan A - 1}{\tan A}$, so $\dfrac{1-\tan A}{1-\cot A} = \dfrac{(1-\tan A)\tan A}{\tan A - 1} = -\tan A$, and its square is $\tan^2 A$.

Both expressions equal $\tan^2 A$.

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